

Verbs and States in Logical Form

A Detailed Explanatory Chapter

Introduction

Verbs are the structural and semantic core of sentences. While nouns introduce entities into discourse, verbs introduce relations, events, processes, and states. A proper logical representation of verbs must therefore capture much more than simple argument structure. It must explain how verbs combine with participants, how they interact with adverbs, how they encode temporal structure, and how they differ in aspectual class.

One of the most influential developments in formal semantics is Davidsonian event semantics, which introduces explicit event variables into logical form. This move allows verbs to denote events, and it provides a uniform attachment point for modifiers such as temporal expressions, adverbs, and instrumental phrases. At the same time, certain verbs — especially stative verbs — behave differently from eventive verbs, which raises fundamental questions about how states should be represented in logical form.

This chapter develops a detailed understanding of how verbs and states are encoded semantically, with careful step-by-step derivations and conceptual explanations.

1. Basic Predicate–Argument Representation

In classical predicate logic, verbs are treated as predicates that take arguments. For example:

Rina laughs

is represented as:

Laugh(r)

Here, the constant r denotes Rina, and the predicate $Laugh(x)$ denotes the property of laughing. This representation captures the basic truth condition: the sentence is true if and only if the individual denoted by r belongs to the set of laughers.

Similarly, a transitive verb:

Rina loves Amit

is represented as:

$Loves(r, a)$

This says that the ordered pair $\langle r, a \rangle$ belongs to the love relation. While this captures relational structure, it does not capture temporal information or modification. For example, if we add an adverb:

Rina laughed loudly

the classical predicate representation does not provide a clear place for “loudly” to attach. This motivates a richer representation.

2. Davidsonian Event Semantics

Davidson proposed that verbs introduce an event variable. Instead of representing “Rina laughed” as $Laugh(r)$, we represent it as:

$\exists e (Laugh(e) \wedge Agent(e, r))$

Let us unpack this carefully.

The variable e ranges over events. The predicate $Laugh(e)$ states that e is a laughing event. The predicate $Agent(e, r)$ states that Rina is the agent (doer) of event e . The existential quantifier asserts that there exists at least one event with these properties.

Why is this useful? Consider:

Rina laughed loudly yesterday.

Now we can represent this as:

$$\exists e(Laugh(e) \wedge Agent(e, r) \wedge Loud(e) \wedge Yesterday(e))$$

Each modifier contributes a conjunct about the same event variable. Loudness and temporal location are properties of the event. This unified representation elegantly captures how modifiers attach.

3. Thematic Roles and Event Structure

Event semantics allows us to explicitly represent participant roles. Consider:

Rina broke the vase.

Logical form:

$$\exists e(Break(e) \wedge Agent(e, r) \wedge Theme(e, v))$$

Let us explain this step-by-step.

- $Break(e)$ says that e is a breaking event.
- $Agent(e, r)$ says Rina performed the event.
- $Theme(e, v)$ says the vase is the entity affected.

This decomposition allows us to represent additional modifiers easily:

Rina broke the vase with a hammer.

$$\exists e(Break(e) \wedge Agent(e, r) \wedge Theme(e, v) \wedge Instrument(e, h))$$

The instrument relation attaches directly to the event. Without event variables, such representations become awkward.

4. Vendler's Aspectual Classes

Verbs differ in how they structure time and change. Vendler identified four major classes: states, activities, accomplishments, and achievements.

4.1 States

States describe static conditions.

Rina knows French.

A simple representation:

$Know(r, french)$

States do not denote dynamic events with internal development. They typically resist progressive marking:

**Rina is knowing French.*

This ungrammaticality reflects semantic incompatibility: the progressive requires dynamic unfolding, which states lack.

4.2 Activities

Activities are dynamic processes without inherent endpoints.

Rina ran.

$\exists e(Run(e) \wedge Agent(e, r))$

Activities combine naturally with duration modifiers:

Rina ran for an hour.

$$\exists e(Run(e) \wedge Agent(e, r) \wedge Duration(e, 1h))$$

The phrase “for an hour” measures the temporal length of the event.

4.3 Accomplishments

Accomplishments involve processes with a natural endpoint.

Rina built a house.

$$\exists e(Build(e) \wedge Agent(e, r) \wedge Theme(e, h))$$

Accomplishments combine naturally with “in an hour”:

Rina built a house in an hour.

The phrase “in an hour” indicates the time taken to reach the endpoint.

4.4 Achievements

Achievements denote instantaneous changes.

Rina won the race.

$$\exists e(Win(e) \wedge Agent(e, r) \wedge Theme(e, race))$$

Achievements resist durational modifiers:

**Rina won the race for an hour.*

Because winning occurs at a single transition point.

5. Tense in Logical Form

Tense situates events relative to speech time.

Rina laughed.

$$\exists e(\text{Laugh}(e) \wedge \text{Agent}(e, r) \wedge \text{Past}(e))$$

Here, $\text{Past}(e)$ indicates that event e occurred before speech time.

Tense may also be represented as a higher-order operator over propositions.

6. Progressive Aspect

The progressive marks ongoingness.

Rina is running.

$$\exists e(\text{Run}(e) \wedge \text{Agent}(e, r) \wedge \text{Ongoing}(e))$$

The progressive highlights internal structure of events.

It conflicts with stative verbs because states lack internal dynamic structure.

7. Perfect Aspect

Rina has left.

$$\exists e(\text{Leave}(e) \wedge \text{Agent}(e, r) \wedge \text{Perfect}(e))$$

The perfect indicates a past event with current relevance.

8. Compositional Derivation Example

Sentence:

Rina broke the vase.

Lexical entry:

$$\text{broke} = \lambda y \lambda x \lambda e (\text{Break}(e) \wedge \text{Agent}(e, x) \wedge \text{Theme}(e, y))$$

Step 1: Apply to object:

$$(\lambda y \dots)(v)$$

$$= \lambda x \lambda e (\text{Break}(e) \wedge \text{Agent}(e, x) \wedge \text{Theme}(e, v))$$

Step 2: Apply to subject:

$$(\lambda x \dots)(r)$$

$$= \lambda e (\text{Break}(e) \wedge \text{Agent}(e, r) \wedge \text{Theme}(e, v))$$

Step 3: Existential closure:

$$\exists e (\text{Break}(e) \wedge \text{Agent}(e, r) \wedge \text{Theme}(e, v))$$

Each step follows function application and beta reduction.

Conclusion

Verbs introduce relational structure and event variables into logical form. Event semantics provides a unified mechanism for representing participants, modifiers, tense, and aspect. States differ fundamentally from events in temporal structure and compatibility with aspectual operators. Vendler's classification reveals systematic aspectual distinctions. Through

event variables and lambda abstraction, formal semantics captures the dynamic structure of verbal meaning with mathematical precision.